ASEAN GEOAI FUSION 2026

PROBLEM-DATA-GeoAI SOLUTION (PDGS) CANVAS

From Real-World Problems to Intelligent Geospatial Solutions

Virtual Hackathon Submission

Project Title: SUNTOWER-AI

Team Name: Bills' ASEAN GeoAI

Team Code: HACK_ID_068

1 PROBLEM STATEMENT

- Thousands of ASEAN telecom towers in off-grid areas still run on costly, high-carbon diesel generators.
- 2026 diesel price shocks are pushing operators (PLDT/Smart, EdgePoint) to solarise — but site choice is largely ad hoc, not data-driven.
- Existing GeoAI/GIS tools are statics – a one-time map that goes stale the moment conditions change
- No systematic, satellite-based way exists to rank which towers deserve solar retrofit first.
- Nothing continuously re-scans for new BTS + solar panel source energy opportunities together or flags whats change
- Result: limited capital misallocated, highest-impact sites missed, slower emissions reduction.

Success: a data-driven ranking that directs capital to the towers with the highest combined cost, emissions and connectivity impact.

2 STAKEHOLDERS & USERS

- **Telecom tower companies** – capital allocation for solar retrofit + new BTS
- **National telecom regulators** – sustainability & USO targets
- **National Utility/Grid Operations** – coal-to-renewable transition planning
- **Off-grid rural communities** – connectivity beneficiaries
- **Renewable energy EPC contractors** – deployment partners
- **Climate/environmental agencies** – emissions reporting

3 SPATIAL CONTEXT

AOI: Off-grid / weak-grid rural regions across ASEAN – outer-island Indonesia, Mindanao (PH), upland Myanmar/Laos

Scale: ASEAN region → Country → Province → individual tower site

Key layers: solar irradiance, land cover/shading, terrain/slope, nighttime lights, population/connectivity-gap zones

4 DATA REQUIREMENTS

Spatial (GEE): Global Solar Atlas PVOUT/GHI (solar yield), Dynamic World V1/Sentinel-2, SRTM 30m DEM, VIIRS nighttime lights DNB, WorldPop 100m, Sentinel-1 SAR, WorldPop, Sentinel-5P

Grid-mix: national utility fuel-max reports (coal vs renewable%) + coal power plant locations, as regional grid-carbon-intensity-proxy

Non-spatial: diesel price & consumption per site, tower opex records, grid-connection status, existing solarisation pipeline

5 GeoAI INTELLIGENT

GeoAI tasks: per-site solar-yield estimation; diesel-cost-avoidance modelling; connectivity-gap overlay scoring; multi-criteria site ranking; automated change-detection between runs; 3 linked ranking models – new-BTS siting, new-PLTS siting, convergence-zone scoring

Outputs: ranked tower priority list, solar-ROI map, emissions-avoided estimate, connectivity-impact score, self-updating priority maps for all 3 interventions, trend-aware forecast

6 GEOAI SOLUTION DESIGN

Solution: SUNTOWER-AI – a satellite-data-driven prioritisation platform for telecom tower solarisation, recommending new BTS, new PLTS, and combined convergence sites

Key modules: solar-yield & shading calculator, diesel-cost/ROI dashboard, connectivity-equity overlay, exportable priority list, scheduled re-scoring pipeline, change-detection & alerting, 3-layer dashboard (BTS / PLTS / combined), coal-vs-renewable grid overlay

7 TECHNOLOGY STACK

GIS: QGIS, Google Earth Engine

Cloud: GCP / AWS

AI/ML: Python, XGBoost, TensorFlow/PyTorch

Data platform: MongoDB

Backend: Node.js/Python

Dashboard: custom-built – React.js + Vite

8 ETHICS, PRIVACY & SUSTAINABILITY

- Use aggregated, operator-level cost data only – avoid unnecessary sensitive commercial data.
- Publish the ranking methodology transparently to prevent prioritisation disputes.
- Guard against deprioritising the hardest-to-reach (often poorest) communities on ROI grounds alone.
- Supports SDG 7 (Clean Energy), SDG 9 (Infrastructure) & SDG 13 (Climate Action).

9 VALUE & IMPACT

Economic: diesel cost avoided; lower exposure to fuel-price shocks

Social: more reliable connectivity in off-grid communities

Environmental: measurable CO_2 avoided per solarised site

KPI: $ diesel avoided/year; tCO_2 avoided; priority sites solarised in Year 1; new priority zones surfaced per update cycle

10 IMPLEMENTATION ROADMAP

Phase 1 (0–3mo): pilot in one off-grid region using open data + a partner operator's tower list

Phase 2 (3–6mo): build & validate ranking model against known solarised vs diesel sites

Phase 3 (6–9mo): pilot dashboard with one operator/regulator for real decisions

Phase 4 (9–15mo): scale to multiple ASEAN countries/operators

Phase 5 (ongoing): refine with post-solarisation performance data

11 FINAL PITCH

SUNTOWER-AI turns satellite solar and land data into a ranked, ASEAN-wide priority list for solarising diesel-dependent telecom towers — cutting costs and emissions while keeping off-grid communities connected, continuously re-scanning satellite data to recommend new BTS sites, new solar (PLTS) sites to displace coal, and where both investments compound, keeping ASEAN's digital and clean-energy planning always current.